

THE TRANSITIVE-ARRAY CONJECTURE FOR THE CLASSICAL YANG–BAXTER EQUATION

ZIJIAN ZENG

ABSTRACT. Berenstein, Greenstein, and Li conjectured that a transitive matrix of colors sends every solution of their transitive classical Yang–Baxter equation to a solution of the ordinary classical Yang–Baxter equation. They reported a verification through matrix size four. We prove the conjecture for every size. The proof is a componentwise expansion: for an ordered triple of direct-sum components, the ordinary Yang–Baxter residual is exactly one of the defining transitive Yang–Baxter relations. An exact finite certificate independently reproduces the reported baseline and checks all 4,573 color-relabeling types at size five.

1. INTRODUCTION

Let C be a set of colors. Berenstein, Greenstein, and Li introduced a transitive version of the classical Yang–Baxter equation (CYBE) and used transitive arrays to construct classical r -matrices [1]. They also formulated a conjecture about transitive square matrices and reported that it had been verified for $n \leq 4$ [1, Conjecture 1.5]. In the first arXiv version the same statement is Conjecture 1.4; the change is only a numbering change.

We prove the conjecture for arbitrary n . The key identity is

$$(1) \quad \text{CYB}_{\mathfrak{g}^{\oplus n}}(\mathbf{r}^{(a)}) = \sum_{j,i,k=1}^n \Phi_{j,i,k} T(a_{ij}, a_{kj}, a_{ki}),$$

where T is the transitive-CYBE residual and $\Phi_{j,i,k}$ places its three tensor factors in the indicated direct-sum components. Matrix transitivity, applied to (k, i, j) , makes every summand vanish. Thus the claimed size-five case is not exceptional.

The proof does not use the finite enumeration. Nevertheless, to audit the index conventions and reproduce the last reported baseline, we provide an exact certificate containing all transitive equality patterns through $n = 5$, modulo color relabeling. A no-import verifier reconstructs the classification, checks every array, and rejects malformed certificates.

2. DEFINITIONS

All vector spaces and Lie algebras are over a fixed field $\mathbb{T}$ of characteristic zero, matching the conventions of [1, Section 2.1]. For a Lie algebra $\mathfrak{g}$ and $x \in \mathfrak{g} \otimes \mathfrak{g}$, the standard leg notation gives $x_{12}, x_{13}, x_{23} \in U(\mathfrak{g})^{\otimes 3}$. Define

$$\text{CYB}_{\mathfrak{g}}(x) = [x_{12}, x_{13}] + [x_{12}, x_{23}] + [x_{13}, x_{23}].$$

Definition 2.1. A family $\mathbf{r} = \{r^{(c)}\}_{c \in C} \subset \mathfrak{g} \otimes \mathfrak{g}$ solves the *transitive CYBE* if

$$(2) \quad T(c, c', c'') := [r_{12}^{(c)}, r_{13}^{(c')}] + [r_{12}^{(c)}, r_{23}^{(c'')}] + [r_{13}^{(c')}, r_{23}^{(c'')}] = 0$$

whenever $c, c', c'' \in C$ and $c' \in \{c, c''\}$.

Definition 2.2. An $n \times n$ matrix $a = (a_{ij})$ with entries in C is *transitive* if

$$(3) \quad a_{ik} \in \{a_{ij}, a_{jk}\} \quad (i, j, k \in [n]).$$

Set $\mathfrak{h} = \mathfrak{g}^{\oplus n}$ and let $\iota_p : \mathfrak{g} \rightarrow \mathfrak{h}$ be inclusion into the p th summand. If $x \in \mathfrak{g} \otimes \mathfrak{g}$, write $x^{p,q} = (\iota_p \otimes \iota_q)(x)$. Formula (1.2) of [1] becomes

$$(4) \quad R = \mathbf{r}^{(a)} := \sum_{i,j=1}^n (r^{(a_{ij})})^{j,i} \in \mathfrak{h} \otimes \mathfrak{h}.$$

3. THE COMPONENT IDENTITY AND THE CONJECTURE

Use the same symbol ι_p for the induced algebra homomorphism $U(\mathfrak{g}) \rightarrow U(\mathfrak{h})$, and define

$$\Phi_{p,q,s} := \iota_p \otimes \iota_q \otimes \iota_s.$$

Lemma 3.1 (Component identity). *For every matrix $a \in C^{n \times n}$ and every family $\{r^{(c)}\}_{c \in C} \subset \mathfrak{g} \otimes \mathfrak{g}$, the element in (4) satisfies*

$$(5) \quad \text{CYB}_{\mathfrak{h}}(R) = \sum_{j,i,k=1}^n \Phi_{j,i,k}(T(a_{ij}, a_{kj}, a_{ki})).$$

Proof. Elements in distinct direct summands commute, so

$$[\iota_p(x), \iota_q(y)] = \begin{cases} \iota_p([x, y]), & p = q, \\ 0, & p \neq q. \end{cases}$$

Expanding the first commutator in $\text{CYB}_{\mathfrak{h}}(R)$, only pairs with equal first-component support survive. Hence

$$[R_{12}, R_{13}] = \sum_{j,i,k=1}^n \Phi_{j,i,k}[r_{12}^{(a_{ij})}, r_{13}^{(a_{kj})}].$$

For the second commutator, the second-component support of the first term must equal the first-component support of the second term, giving

$$[R_{12}, R_{23}] = \sum_{j,i,k=1}^n \Phi_{j,i,k}[r_{12}^{(a_{ij})}, r_{23}^{(a_{ki})}].$$

Finally, equal third-component supports give

$$[R_{13}, R_{23}] = \sum_{j,i,k=1}^n \Phi_{j,i,k} [r_{13}^{(a_{kj})}, r_{23}^{(a_{ki})}].$$

Their sum is (5). $\square$

Theorem 3.2. *Let $n \geq 1$, let $a \in C^{n \times n}$ be transitive, and let $\mathbf{r} = \{r^{(c)}\}_{c \in C}$ solve the transitive CYBE. Then $\mathbf{r}^{(a)}$ solves the ordinary CYBE in $\mathfrak{g}^{\oplus n}$. In particular, Conjecture 1.4 of arXiv:2602.02342v1 (Conjecture 1.5 of v2) holds for $n = 5$ and, more generally, for all n .*

Proof. For each $j, i, k \in [n]$, apply (3) to the ordered triple (k, i, j) . It gives

$$a_{kj} \in \{a_{ki}, a_{ij}\}.$$

Therefore the transitive-CYBE relation gives $T(a_{ij}, a_{kj}, a_{ki}) = 0$. Every summand in Lemma 3.1 vanishes. $\square$

Remark 3.3 (Degeneracies). No indices are required to be distinct. If labels among j, i, k coincide, the three copies of $U(\mathfrak{h})$ in $U(\mathfrak{h})^{\otimes 3}$ remain separate, and $\Phi_{j,i,k}$ is still defined. The proof does not use linear independence of the images of these maps: each input is zero before it is embedded. Thus constant arrays, repeated colors, the zero family, and all diagonal cases are covered. An infinite color set causes no issue because (4) uses only finitely many colors.

4. EXACT FINITE AUDIT AT SIZE FIVE

The finite audit quotients color relabeling but deliberately retains index permutations. Proposition 3.14 of [1] classifies the equality patterns: the two-color case is reconstructed from a set partition of $[n]$, a permutation of its blocks, and a diagonal bit on each block; every pattern using at least three colors is row-constant or column-constant on a set partition. The resulting exact counts are shown in Table 1.

n	1 col.	2 col.	3 col.	4 col.	5 col.	total	components
2	1	4	0	0	0	5	40
3	1	36	2	0	0	39	1,053
4	1	364	12	2	0	379	24,256
5	1	4,500	50	20	2	4,573	571,625

TABLE 1. Canonical equality patterns and checked ordered residual components.

The serialized certificate stores every canonical pattern. An independent Python verifier does not import the generator. It strictly checks the schema, reconstructs the complete sets, tests transitivity, and verifies the middle-color condition in every ordered residual component. A second program

starts from all ordered pairs of the n^2 terms in R and reconstructs the three commutator expansions of Lemma 3.1. The combined audit covers 4,996 patterns and 596,974 residual components for $2 \leq n \leq 5$. All arithmetic is discrete; no floating-point or modular calculation is used.

For a separate falsification route, we used the integral $\mathfrak{sl}_2$ family

$$r^{(0)} = H \otimes H + 4E \otimes F, \quad r^{(1)} = -H \otimes H - 4F \otimes E,$$

where $[H, E] = 2E$, $[H, F] = -2F$, and $[E, F] = H$. Exact tensor evaluation on all 4,501 one- and two-color size-five patterns yielded zero in 562,625 ordered components. This is diagnostic negative evidence only; the proof of Theorem 3.2 is the component identity.

The certificate and verifier are included with the source. From the project root, the complete verification command is

`make verify`

DATA, CODE, AND COMPUTATIONAL DISCLOSURE

Computer assistance was used to search the current literature, reproduce the finite baseline, check the symbolic support expansion, test an exact $\mathfrak{sl}_2$ family, audit citations and L^AT_EX, and prepare the manuscript. Generative AI assistance was used in proof discovery, code, and drafting. The mathematical proof is the explicit identity (5) and does not depend on software output. No proof assistant such as Lean, Coq, or Isabelle was used.

REFERENCES

1. Arkady Berenstein, Jacob Greenstein, and Jian-Rong Li, *Monomial bialgebras*, 2026, [arXiv:2602.02342v2 \[math.QA\]](#), revised February 8, 2026.

INSTITUTE OF COMPUTER SCIENCE AND DIGITAL INNOVATION, UCSI UNIVERSITY,
KUALA LUMPUR, 56000, MALAYSIA

Email address: `zijianzeng@foxmail.com`

Email address: `1002266693@ucsiuniversity.edu.my`